

Ta-I Wu

Taiwan | 0978565941 | juniorwu1017@gmail.com

Education

National Cheng Kung University

Master of Science in Computer Science and Information Engineering
IIR Lab | Advisor: Professor Jung-Hsien Chiang

Tainan, Taiwan

2026 - Present

National Taiwan University

Bachelor of Science in Agricultural Chemistry
EnvNano Lab | Advisor: Professor Yang-Hsin Shih

Taipei, Taiwan

2020 - 2025

Relevant Coursework

Computer Science & Data: Introduction to Computer and Programming; Computer Programming in Python (A+); Data Structures; Operating Systems; Statistics; Data Analysis and Machine Learning with Python (A+); Web Application Programming (A+); Principles of Financial Computing (graduate-level CSIE course).

Chemistry: Analytical, Physical, Organic, General, Biochemistry, and Soil Chemistry, including laboratory coursework.

Selected Projects

Car Insurance Claim Prediction | Machine Learning / Deep Learning

- Developed a Python-based workflow for data preprocessing, model development, and performance evaluation to predict whether a policyholder would file a claim within the next six months.
- Applied statistical analysis and visualization to strengthen data-driven problem solving relevant to process monitoring and yield improvement.

Online Bookstore | Full-Stack Web Application

- Built front-end, back-end, and database components for an online bookstore, gaining end-to-end web application development experience.

Research Experience

Graduation Project: RB5 Photodegradation Using g-C₃N₄

- Synthesized g-C₃N₄ photocatalysts from melamine / urea precursors and compared secondary-calcination and ammonium-chloride doping conditions to establish a repeatable preparation workflow.
- Evaluated RB5 degradation under 420 nm visible light and analyzed UV-Vis, XRD, FTIR, photoluminescence, particle sedimentation, and Z-average data to link process conditions with material performance.

Industry-Academic Project: Oil-Contaminated Water Analysis | EnvNano Lab

- Prepared aqueous and oil-contaminated samples, observed mixing and phase stability, compared controlled conditions, and maintained structured experimental records for root-cause-oriented troubleshooting.

Skills

Programming & Data: C, Python, data cleaning and visualization, statistical analysis, machine learning, deep learning fundamentals, and spreadsheet tools.

Web Development: Front-end development, back-end development, and database integration.

Laboratory & Materials: Material synthesis, calcination, solution and environmental sample preparation, process control, reproducibility assessment, XRD, FTIR, UV-Vis, photoluminescence, particle sedimentation, and Z-average analysis; familiarity with XPS and electrochemical workflows.

Languages: Mandarin Chinese (native); English (TOEIC 785; technical reading and working proficiency).